Overview

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Autono-MaaS Business Logic

Seneschal: https://seneschal.base44.app ・・・ Demand-generation implementation example
Federation: https://federation.base44.app ・・・ Urban-operation packaging study framework

The concepts presented in this portfolio are not proposals for implementation. They demonstrate my concrete understanding of the strategy and execution requirements needed to make autonomous-mobility businesses commercially viable.

The success of Autono-MaaS is determined not by autonomous driving technology itself, but by a business structure that converts operation into profit. As driver labor cost decreases through automation, the dominant cost factor converges toward energy cost, while vehicle utilization is required to recover CAPEX and load factor is required to increase revenue per trip. Seneschal smooths demand fluctuations by matching customer availability with mobility and services, while motion-sickness-free vehicle control transforms travel time into work, rest, and service value.

Because vehicle-grade quality requirements are strict, next-generation vehicle technologies should first be developed for stationary applications, then installed into service vehicles as they become ready. Service vehicles can absorb CAPEX through high utilization and high load factor, enabling a DevOps-style development model unlike conventional mass-production vehicles. Federation increases vehicle utilization through time-based service exchange and ridesharing, and increases load factor through passenger-freight integration and collaborative logistics, while planned mobility schedules optimize PPA energy management and enable AVP-integrated fueling to establish a carbon-neutral energy platform for autonomous mobility. Autono-MaaS is an urban operating system that controls demand, vehicles, energy, and infrastructure as one system to convert utilization, load factor, and energy cost into profit.

PPA Service for Autonomous Mobility Foundation

As autonomous driving reduces driver labor cost, energy becomes the dominant operating-cost factor. The PPA service therefore establishes structural energy-cost competitiveness for Autono-MaaS through distributed energy infrastructure.

Corporate PPA services support Scope 3 decarbonization while establishing distributed eVTOL operating hubs. Idle corporate vehicles are shared for employee commuting to improve fleet utilization and reduce Scope 3 transportation emissions. Residential trigeneration services establish distributed neighborhood mobility hubs, providing the node density required for daily autonomous mobility services.

Blockchain-enabled P2P energy and mobility sharing connects corporate and residential nodes into a shared mobility and energy ecosystem, while renewable electricity, water-pipeline micro-hydropower, magnetic refrigeration, hydrogen production, superconducting power cables, liquid-hydrogen pipelines, and liquid-hydrogen-cooled HPC establish the infrastructure foundation for infrastructure-coordinated autonomy and continuous ODD expansion.

Infrastructure Layer	Control Variable	Target	Business Purpose
Renewable PPA	Renewable supply ratio	Maximize	Low-cost renewable electricity
Water Infrastructure	Micro-hydropower output	Maximize	Distributed baseload generation
Hydrogen Production	Electrolysis utilization	Maximize	Hydrogen production
Cooling	Magnetic refrigeration efficiency	Maximize	LH_2 production & HPC cooling
Energy Network	Superconducting cable utilization	≥80%	Inter-node power sharing
Hydrogen Network	LH_2 pipeline utilization	≥70%	Hydrogen distribution
HPC Node	AI compute availability	≥99.9%	Infrastructure AI computing
Infrastructure Coordination	Infrastructure coverage	Expand	Vehicle–infrastructure cooperation
Autonomous Mobility	ODD coverage	Expand	City-wide autonomous operation

AVP-to-MaaS Execution

This concept begins with an existing OEM customer use case: enabling refueling to be completed while the user is shopping. → A: time control

By integrating infrastructure-coordinated automated valet parking (AVP) with fueling, vehicles autonomously park, refuel, and return within dwell time, eliminating dedicated refueling stops. → B: time + location control

As autonomous driving progresses, parking demand shifts, creating underutilized parking spaces repurposed as operational nodes. By leveraging these assets, dual-use infrastructure enables daytime refueling for private users and nighttime fleet operations. → C: time allocation control

Liquid-hydrogen-cooled HPC nodes provide compute capacity for infrastructure-coordinated autonomy. Any ODD expansion remains subject to scenario coverage, V&V evidence, a validated safety case, and regulatory approval.

Applied to large-scale parking operators, operational nodes exchange modular service containers according to time-of-day demand, balancing utilization across mobility, logistics, retail, and public services. This transforms Node into a MaaS foundation that mitigates demand fluctuations and increases asset utilization. → D: operational control

Extending this logic to public roads evolves the system into a city-wide mobility platform. → E: flow control

Through autonomous driving, vehicles synchronize in time and space, forming organized carrier operations rather than independent movements. → F/G: synchronization / platoon control

This expansion to public roads and coordinated carrier systems enables a new transportation system. Because this system is built on coordinated flows and shared infrastructure, Autonomous mobility expands from vehicle operation into integrated mobility and infrastructure coordination. → H: system-level control

Logic	Layer	KPI	Formula	Value	Unit
A	Demand	Demand generation rate	generated demand ÷ total potential demand	≥0.30	ratio
A	Demand	Conversion rate	booking ÷ request	≥0.23	ratio
B	Demand	Matching rate	matched ÷ request	≥0.90	ratio
D	Demand	Trip request rate	executed trips ÷ matched	≥0.95	ratio
D	Operation	Dispatch success rate	completed dispatch ÷ dispatch request	≥0.95	ratio
D	Operation	Dispatch response time	total dispatch response time ÷ dispatches	≤60	sec
D	Operation	Utilization	operating time ÷ total time	≥0.75	ratio
D	Operation	Fleet idle ratio	idle vehicles ÷ total vehicles	≤0.25	ratio
F	Vehicle	Latency	response time	≤1	sec
F	Vehicle	Control delay	actuation delay	≤20	ms
F	Vehicle	Planning latency	planning delay	≤100	ms
G	Vehicle	Control stability	spacing variance	≤5	%
E	Infrastructure	Throughput	vehicles ÷ time	≥40	veh/h
C	Infrastructure	Node utilization	used nodes ÷ total nodes	≥0.80	ratio
F	Infrastructure	Node latency	node delay	≤50	ms
H	Infrastructure	Energy cost per trip	energy cost ÷ trips	≤¥500	JPY/trip